Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A projectile is fired from the surface of the earth with a velocity of $5\ \mathrm{ms}^{-1}$ and angle θ with the horizontal. Another projectile fired from another planet with a velocity of $3\ \mathrm{ms}^{-1}$ at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in ms^{-2}) given $\mathrm{g} = 9.8\ \mathrm{m/s}^2$

(A) 3.5

(B) 5.9

(C) 16.3

(D) 110.8

Q.2

A projectile is thrown at an angle of 40° with the horizontal and its range is R_1. Another projectile is thrown at an angle 40° with the vertical and its range is R_2. What is the relation between R_1 and R_2?

(A) $\mathrm{R}_1 = \mathrm{R}_2$

(B) $\mathrm{R}_1 = 2\mathrm{R}_2$

(C) $2\mathrm{R}_1 = \mathrm{R}_2$

(D) $\mathrm{R}_1 = 4\mathrm{R}_2/5$

Q.3

A particle is projected at an angle of elevation α and after t seconds it appears to have an angle of elevation β as seen from point of projection. The initial velocity will be

(A) $\frac{gt}{2\sin(\alpha-\beta)}$

(B) $\frac{gt\cos\beta}{2\sin(\alpha-\beta)}$

(C) $\frac{\sin(\alpha-\beta)}{2\ gt}$

(D) $\frac{2\sin(\alpha-\beta)}{gt\cos\beta}$

Q.4

If a particle is projected with speed u from ground at an angle with horizontal ,then radius of curvature of a point where velocity vector is perpendicular to initial velocity vector is given by

(A) $\frac{u^2\cos^2\theta}{g}$

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(B) $\frac{u^2 \cot^2 \theta}{g \sin \theta}$

(C) $\frac{u^2}{g}$

(D) $\frac{u^2 \tan^2 \theta}{g \cos \theta}$

Q.5

A ball rolls off to the top of a staircase with a horizontal velocity $\mathrm{um/s}$. If the steps are h metre high and b metre wide, the ball will hit the edge of the nth step, if

(A) $\mathrm{n}=\frac{2\mathrm{hu}}{\mathrm{gb}^2}$

(B) $n=\frac{2hu^2}{gb}$

(C) $n=\frac{2hu^2}{gb^2}$

(D) $\mathrm{n}=\frac{\mathrm{hu}^2}{\mathrm{gb}^2}$

Q.6

A body of mass m is projected horizontally with a velocity v from the top of a tower of height h and it reaches the ground at a distance x from the foot of the tower. If a second body of mass $2\ \mathrm{m}$ is projected horizontally from the top of a tower of height $2\ \mathrm{h}$, it reaches the ground at a distance $2\mathrm{x}$ from the foot of the tower. The horizontal velocity of the second body is

(A) v

(B) $2v$

(C) $\sqrt{2v}$

(D) $v/2$

Q.7

The velocity of projection of oblique projectile is $(6\hat{\mathrm{i}}+8\hat{\mathrm{j}})\mathrm{ms}^{-1}$. The horizontal range of the projectile is

(A) $4.9\ \mathrm{m}$

(B) $9.6\ \mathrm{m}$

(C) $19.6\ \mathrm{m}$

(D) $14\ \mathrm{m}$

Q.8

A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity u at an angle $\left(\frac{\pi}{2}-\theta\right)$ with the horizontal, it reaches maximum height H_2. The relation between the horizontal range R of the projectile, heights H_1 and H_2 is

(A) $\mathrm{R}=4\sqrt{\mathrm{H}_1\mathrm{H}_2}$

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(B) $\mathrm{R}=4\left(\mathrm{H}_1-\mathrm{H}_2\right)$

(C) $\mathrm{R}=4\left(\mathrm{H}_1+\mathrm{H}_2\right)$

(D) $\mathrm{R}=\frac{\mathrm{H}_1^2}{\mathrm{H}_2^2}$

Q.9

A body is projected with a speed u at an angle α with the horizontal. The speed of the body when its velocity vector makes an angle β with the vertical is

(A) $\frac{u \cos \alpha}{\cos \beta}$

(B) $\frac{u \cos \alpha}{\sin \beta}$

(C) $\frac{u \sin \alpha}{\cos \beta}$

(D) $\frac{u \sin \alpha}{\sin \beta}$

Q.10

An artillary piece which consistently shoots its shells with the same muzzle speed has a maximum range R. To hit a target which is $\mathrm{R}/2$ from the gun and on the same level, then what will be the elevation angle (in degree) of the gun?

(A) 10

(B) 18

(C) 12

(D) 15

Q.11

A projectile is thrown in the upward direction making an angle of 60° with the horizontal direction with a velocity of $147 \mathrm{~m~s}^{-1}$. Then what will be the time (in sec) after which its inclination with the horizontal is 45° ? (Use g = 9.8 $\mathrm{m/s}^2$)

(A) 5.59

(B) 5.39

(C) 5.69

(D) 5.49

Q.12

A body is projected horizontally with a speed u from the top A of an plane ABC inclined at an angle $\theta=30^{\circ}$ with the horizontal. It strikes the plane at point P. The distance AP is given by

(A) $\frac{u^2}{2g}$

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(B) $\frac{3}{2}\frac{u^2}{g}$

(C) $\frac{2}{3}\frac{u^2}{g}$

(D) $\frac{4}{3}\frac{u^2}{g}$

Q.13

A body is projected horizontally from the top of a building. It strikes the ground after a time t with its velocity vector making an angle θ with the horizontal. The speed with which the body is projected is

(A) $\frac{gt}{\sin\theta}$

(B) $\frac{gt}{\cos\theta}$

(C) $\frac{gt}{\tan\theta}$

(D) $\frac{gt}{\cot\theta}$

Q.14

A building P is $150\ \mathrm{m}$ tall and is at a distance of $100\ \mathrm{m}$ from another building Q. A ball is thrown horizontally from the top of building P such that it enters a window in building Q. The height of the window above the ground is $25\ \mathrm{m}$. The speed (in ms^{-1}) with which the ball is thrown must be (take $g = 10\ \mathrm{ms}^{-2}$).

(A) 20

(B) 30

(C) 40

(D) 50

Q.15

Two bodies are projected horizontally in opposite directions with velocities u_1 and u_2 from the top of a building. If their velocitices become perpendicular to each other after falling through a distance h, then

(A) $h = \sqrt{2}\frac{u_1 u_2}{g}$

(B) $h = \frac{2u_1 u_2}{g}$

(C) $h = \frac{u_1 u_2}{\sqrt{2}g}$

(D) $h = \frac{u_1 u_2}{2g}$

Q.16

It is possible to project a particle with a given velocity in two possible ways so as to make it pass through a point P at a distant r from the point of projection. The product of the times taken to reach this point in the two possible ways is then proportional to

(A) $1/r$

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(B) r

(C) r^3

(D) $\frac{1}{r^2}$

Q.17

From the top of a tower of height $40\ \mathrm{m}$, a ball is projected upwards with a speed of $20\ \mathrm{ms}^{-1}$ at an angle of 30° to the horizontal. If $g = 10\ \mathrm{ms}^{-2}$, after how long will the ball hit the ground?

(A) $1\ \mathrm{s}$

(B) $2\ \mathrm{s}$

(C) $3\ \mathrm{s}$

(D) $4\ \mathrm{s}$

Q.18

A body is projected with a velocity u at an angle θ with the horizontal. The velocity of the body will become perpendicular to the velocity of projection after a time t given by

(A) $\frac{2u\sin\theta}{g}$

(B) $\frac{u\sin\theta}{g}$

(C) $\frac{2u}{g\sin\theta}$

(D) $\frac{u}{g\sin\theta}$

Q.19

A body is projected from the ground with a speed u at an angle θ with the horizontal. When the body is at the highest point of the trajectory, its speed is $\sqrt{2/3}$ times its speed when it is at half the maximum height. The value of θ is

(A) 30°

(B) 45°

(C) 60°

(D) $\sin^{-1}\left(\frac{2}{3}\right)$

Q.20

A body is projected from the ground with a velocity u at an angle θ with the horizontal. The average velocity of the body between its point of projection and the highest point of its trajectory is

(A) $\frac{u}{2}(1+\cos\theta)$

(B) $\frac{u}{2}\left(1+\cos^2\theta\right)^{1/2}$

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(C) $\frac{u}{2}\left(1+2\cos^2\theta\right)^{1/2}$

(D) $\frac{u}{2}\left(1+3\cos^2\theta\right)^{1/2}$

Answers & Solutions

Q.1 Answer:

3.5

Solution:

Horizontal range $= \frac{\mathrm{u}^2 \sin 2\theta}{\mathrm{g}}$ sog $\propto \mathrm{u}^2$

or $\frac{g_{\text{planet}}}{g_{\text{earth}}} = \frac{\left(u_{\text{planet}}\right)^2}{\left(u_{\text{earth}}\right)^2}$

Therefore $g_{\text{planet}} = \left(\frac{3}{5}\right)^2 \left(9.8\ \mathrm{m/s^2}\right)$

$= 3.5\ \mathrm{m/s^2}$

Q.2 Answer:

$\mathrm{R}_1 = \mathrm{R}_2$

Solution:

R is same for both θ and $(90^\circ - \theta)$. If angle w.r.t. vertical is 40° then w.r.t. horizontal direction it will be $90^\circ - 40^\circ = 50^\circ$

Q.3 Answer:

$\frac{gt\cos\beta}{2\sin(\alpha-\beta)}$

Solution:

$t = \frac{2u\sin(\alpha-\beta)}{g\cos\beta}$.; $u = \frac{gt\cos\beta}{2\sin(\alpha-\beta)}$

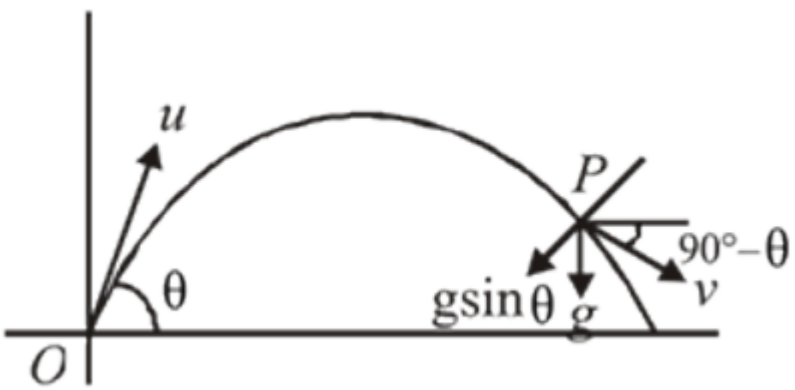
Q.4 Answer:

$\frac{u^2\cot^2\theta}{g\sin\theta}$

Solution:

Horizontal components of velocity at O and are equal. $\therefore v\cos(90^\circ-\theta) = u\cos\theta$

$\frac{u^2\cot^2\theta}{g\sin\theta} = R$



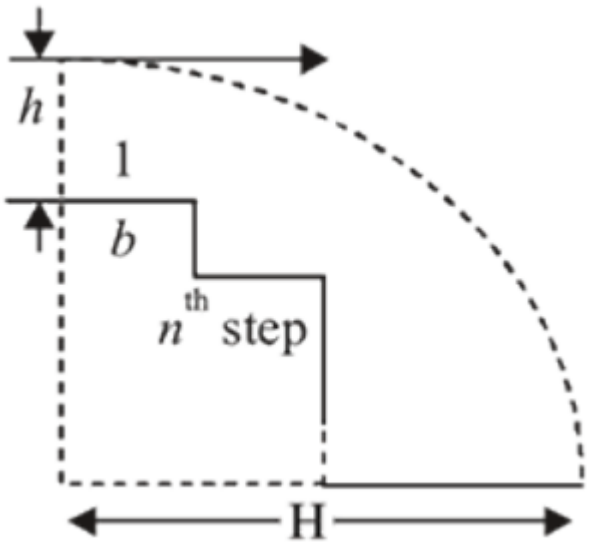
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Q.5 Answer:

$n = \frac{2hu^2}{gb^2}$

Solution:

If the ball hits the nth step, then the horizontal distance traversed $= nh$. Here, the velocity along the horizontal direction $= u$. Initial velocity along the vertical direction $= 0$



So $nb = ut\ nh = 0 + \frac{1}{2}gt^2$

From $t = \frac{nb}{v}$**, putting in eq. (ii)** $nh = \frac{1}{2}g \times \left(\frac{nb}{u}\right)^2$ **or** $n = \frac{2hu^2}{gb^2}$

Q.6 Answer:

$\sqrt{2v}$

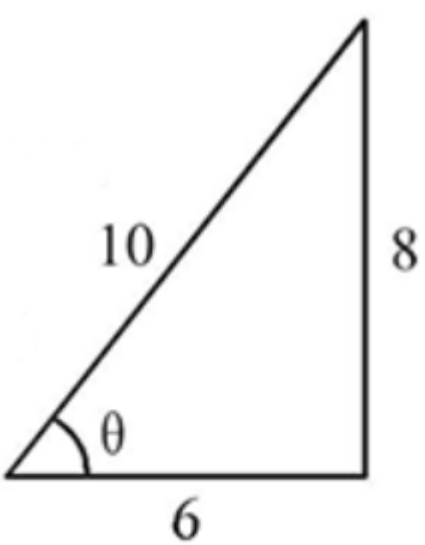
Solution:

$x = v\sqrt{\frac{2h}{g}}, 2x = v'\sqrt{\frac{2(2h)}{g}} = \sqrt{2}v$

Q.7 Answer:

9.6 m

Solution:



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$\sin\theta = \frac{8}{10}$
$\cos\theta = \frac{6}{10}$
$\mathrm{R} = \frac{\mathrm{v}^2 \sin 2\theta}{\mathrm{g}} = \frac{2\mathrm{v}^2 \sin\theta\cos\theta}{\mathrm{g}}$
$\mathrm{R} = 2 \times 10 \times 10 \times \frac{8}{10} \times \frac{6}{10} \times \frac{1}{10} = 9.6\ \mathrm{m}$

Q.8 Answer:

$\mathrm{R} = 4\sqrt{\mathrm{H_1 H_2}}$

Solution:

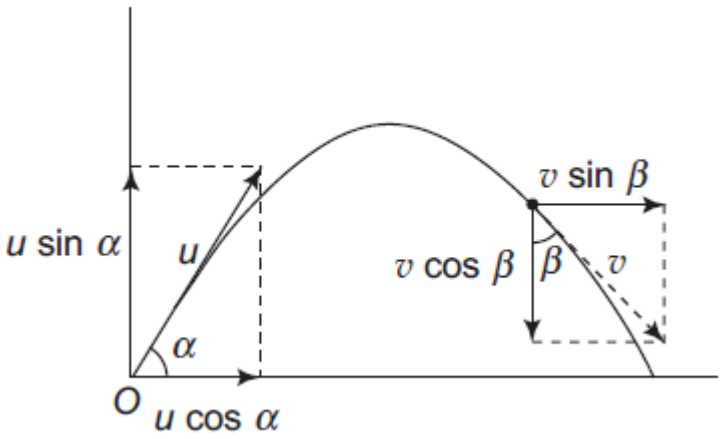
$\mathrm{H_1} = \frac{\mathrm{u}^2 \sin^2\theta}{2\,\mathrm{g}}$
and $\mathrm{H_2} = \frac{\mathrm{u}^2 \sin^2(90^\circ - \theta)}{2\,\mathrm{g}} = \frac{\mathrm{u}^2 \cos^2\theta}{2\,\mathrm{g}}$
$H_1 H_2 = \frac{u^2 \sin^2\theta}{2\,\mathrm{g}} \times \frac{u^2 \cos^2\theta}{2\,\mathrm{g}} = \frac{\left(u^2 \sin 2\theta\right)^2}{16 g^2} = \frac{R^2}{16}$
$\therefore R = 4\sqrt{\mathrm{H_1 H_2}}$

Q.9 Answer:

$\frac{u\cos\alpha}{\sin\beta}$

Solution:

Let the speed of the body be v when its velocity vector makes an angle β with the vertical (Fig.) Horizontal component of u is $u\cos\alpha$ and the horizontal component of v is $v\sin\beta$. Since the horizontal component of velocity remains constant, $\therefore \quad u\cos\alpha = v\sin\beta$
$\Rightarrow \quad v = \frac{u\cos\alpha}{\sin\beta}$**, which is choice**



Q.10 Answer:

15

Solution:

Range of a projectile is maximum when it is projected at an angle of 45° and is given by $R_{\max} = \frac{u^2}{g}$, where u is the velocity of projection $\Rightarrow R = \frac{u^2}{g} \quad \therefore u^2 = Rg \quad \ldots$ **(i)**
Now, to hit a target at a distance $(\mathrm{R}/2)$ from the gun, we must have $\frac{R}{2} = \frac{u^2 \sin 2\theta}{g}$**, where θ is the angle of projection.** $\Rightarrow \frac{R}{2} = \frac{Rg \sin 2\theta}{g}$**; from (i)**

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$\Rightarrow \sin 2\theta = \frac{1}{2} \Rightarrow \sin 2\theta = \sin 30^\circ$
$\Rightarrow 2\theta = 30^\circ \quad \therefore \theta = 15^\circ$

Q.11 Answer:

5.49

Solution:

At the two points of the trajectory during projection, the horizontal component of the velocity is the same.

$\Rightarrow \quad u\cos 60^\circ = v\cos 45^\circ$

$\Rightarrow \quad 147 \times \frac{1}{2} = v \times \frac{1}{\sqrt{2}} \Rightarrow \frac{147}{\sqrt{2}}\ \mathrm{m/s}$

Vertical component of $\mathrm{u} = \mathrm{u}\sin 60^\circ$

$$= \frac{147\sqrt{3}}{2}\ \mathrm{m}$$

Vertical component of $\mathrm{v} = \mathrm{v}\sin 45^\circ$

$$= \frac{147}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{147}{2}\ \mathrm{m}$$

$$\text{but } \mathrm{v_y} = \mathrm{u_y} + \mathrm{a_y t} \Rightarrow \frac{147}{2} = \frac{147\sqrt{3}}{2} - 9.8\mathrm{t}$$

$$\Rightarrow 9.8\mathrm{t} = \frac{147}{2}(\sqrt{3} - 1) \Rightarrow \mathrm{t} = 5.49\ \mathrm{s}$$

Q.12 Answer:

$\frac{4}{3}\frac{u^2}{g}$

Solution:

If the time taken by the body to strike the plane at P **is** t **(Fig.), then the horizontal and vertical distances travelled by it are**

$x = ut$ **(i)**

and $y = \frac{1}{2}gt^2$ **(ii)**

Eliminating t **from (i) and and (ii),** $y = \frac{gx^2}{2u^2}$

Now $\tan\theta = \frac{AD}{DP} = \frac{y}{x}$

$\Rightarrow \quad y = x\tan\theta$. Thus

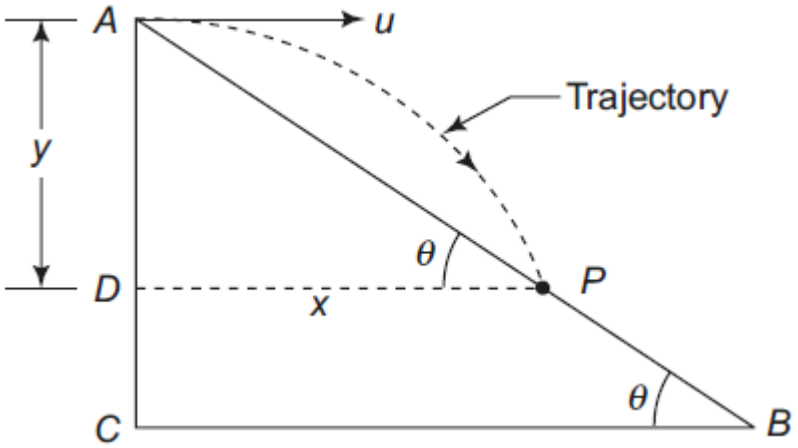
$\Rightarrow x\tan\theta = \frac{gx^2}{2u^2}$

$\Rightarrow \quad x = \frac{2u^2\tan\theta}{g}$

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$$\text{Distance } AP = \sqrt{x^2 + y^2}$$
$$= \sqrt{x^2 + x^2 \tan^2\theta}$$
$$= x\sqrt{1 + \tan^2\theta}$$
$$= x \sec\theta$$
$$= \frac{2u^2 \tan\theta \sec\theta}{g}$$
$$\text{or AP} = \frac{2u^2 \sin\theta}{g\cos^2\theta}$$

For $\theta = 30^\circ$, $AP = \frac{2u^2 \sin 30^\circ}{g \cos^2 30^\circ} = \frac{4u^2}{3g}$



Q.13 Answer:

$\frac{gt}{\tan\theta}$

Solution:

Let u **be the speed with which the body is projected and let** v_x **and** v_y **be the horizontal and vertical components of the velocity at time** t**. Then**

$\tan\theta = \frac{v_y}{v_x} = \frac{v_y}{u} \quad (\because v_x = u \text{ remains constant })$

$\Rightarrow u = \frac{v_y}{\tan\theta}$

Now, since the initial y **component** (u_y) **is zero, we have**

$-v_y = 0 - gt$

$\Rightarrow v_y = gt$

Using (ii) in (i), we get $u = \frac{gt}{\tan\theta}$**, which is choice (c)**

Q.14 Answer:

20

Solution:

It follows from the figure that the horizontal range is $R = 100$ m **and the vertical displacement** $h = (150 - 25) = 125$ m

Time taken to fall through this distance is

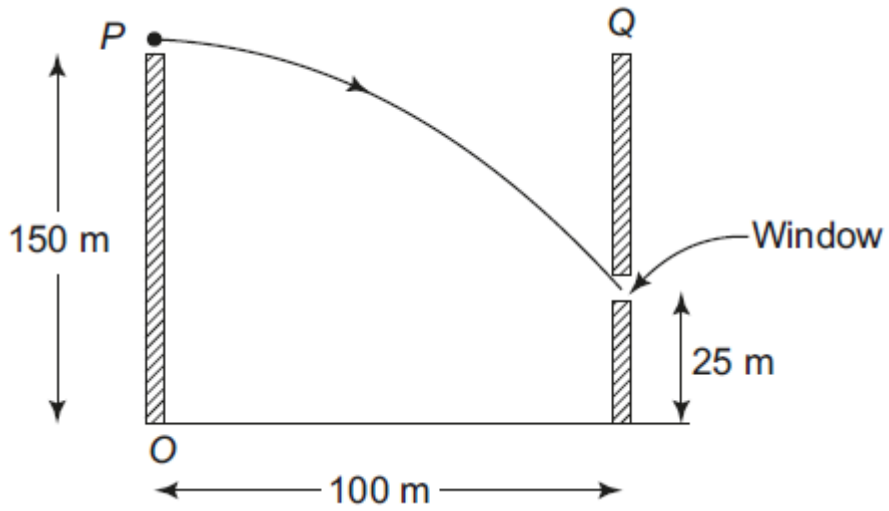
Now

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$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times125}{10}} = 5\ \text{s}$

$R = ut$

$\Rightarrow \quad u = \frac{R}{t} = \frac{100}{5} = 20\ \text{m s}^{-1}$



Q.15 Answer:

$h = \frac{u_1 u_2}{2g}$

Solution:

Let u_1 be along the positive x-axis and u_2 along the negative x-axis. Their velocities at any time t are

$v_1 = u_1\hat{\text{i}} + (gt)\hat{\text{j}}$

$v_2 = -u_2\hat{\text{i}} + (gt)\hat{\text{j}}$

Let t be the time at which v_1 becomes perpendicular to v_2, then

$\therefore h = \frac{1}{2}gt^2 = \frac{1}{2}g \times \frac{u_1 u_2}{g^2} = \frac{u_1 u_2}{2g}$

Q.16 Answer:

r

Solution:

The range of a projectile is $r = 2v_0^2 \cos\theta \sin\theta / g$, the two possible angles of projection are θ and $(90^\circ - \theta)$. The time of flight corresponding to these two angles are

$t_1 = \frac{2v_0 \sin\theta}{g}$ and

$t_2 = \frac{2v_0 \sin(90^\circ - \theta)}{g} = \frac{2v_0 \cos\theta}{g}$

so that $t_1 t_2 = \frac{4v_0^2 \sin\theta \cos\theta}{g^2} = \frac{2r}{g}$. Thus $t_1 t_2 \propto r$.

Hence the correct choice is (b).

Q.17 Answer:

$4\ \text{s}$

Solution:

The horizontal range $R = AC$ is

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$$R = \frac{u^2 \sin 2\theta}{g} = \frac{20 \times 20 \times \sin 60^\circ}{10}$$
$$= 20\sqrt{3}\ \text{m}$$

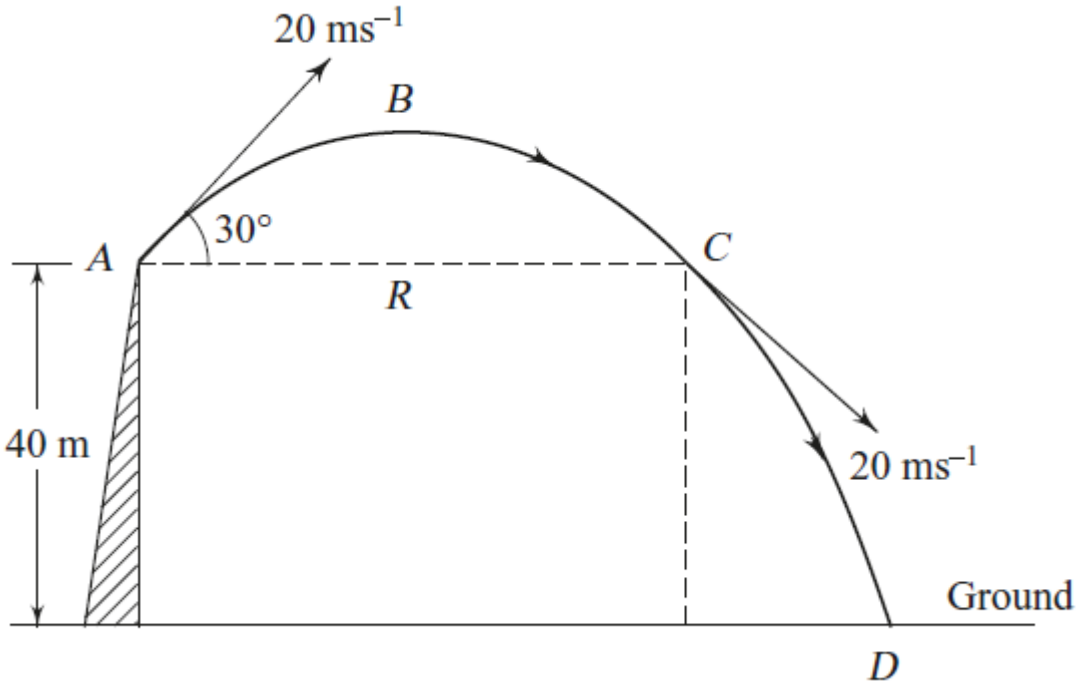
At C, the velocity of the ball is again $20\ \text{ms}^{-1}$ directed down as shown in Fig.
The downward vertical component of this velocity is $20 \sin 30^\circ = 10\ \text{ms}^{-1}$. The ball will hit the ground at D after travelling a vertical distance $h = 40$ m. If t_1 is the time taken for this, then

$h = ut_1 + \frac{1}{2}gt_1^2$
or $40 = 10t_1 + 5t_1^2$
or $t_1^2 + 2t_1 - 8 = 0$

The positive root of this quadratic equation is $t_1 = 2$ s. Now, the time of flight from A to C via B is

$$t_2 = \frac{2u \sin\theta}{g} = \frac{2 \times (20) \times \sin 30^\circ}{10}$$
$$= \frac{2 \times 20 \times \sin 30^\circ}{10} = 2\ \text{s}$$

$\therefore$ Total time taken $= t_1 + t_2 = 2\ \text{s} + 2\ \text{s} = 4$ s. Hence the correct choice is (d).



Q.18 Answer:

$\frac{u}{g \sin\theta}$

Solution:

Velocity of projection is $\text{v}_0 = (u\cos\theta)\hat{\text{i}} + (u\sin\theta)\hat{\text{j}}$
At time t, the velocity of the body is
$\text{v} = (u\cos\theta)\hat{\text{i}} + (u\sin\theta - gt)\hat{\text{j}}$
The dot product of v_0 and v is
$\text{v}_0 \cdot \text{v} = u^2\cos^2\theta + u\sin\theta(u\sin\theta - gt)$
or $\text{v}_0 \cdot \text{v} = u^2 - (u\sin\theta)gt \quad$ (i)
Since v is perpendicular to v_0, $\text{v}_0 \cdot \text{v} = 0$. Using this in (i), we have
$0 = u^2 - (u\sin\theta)gt \quad$ or $t = \frac{u}{g\sin\theta}$

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Hence the correct choice is (d).

Q.19 Answer:

45°

Solution:

$h_{\max} = \frac{u^2 \sin^2\theta}{2g} \Rightarrow gh_{\max} = \frac{u^2 \sin^2\theta}{2}$

At the highest point, the speed is

$$\begin{aligned} v_x &= u\cos\theta \\ \Rightarrow v_x^2 &= u^2\cos^2\theta. \quad (1) \end{aligned}$$

The speed of the body at half the maximum height is

$v^2 = u^2 - 2g \times \frac{h_{\max}}{2} = u^2 - gh_{\max}$ **(2)**

It is given that $v_x = v \times \sqrt{\frac{2}{3}}$**. Substituting (1) and (2) in this equation, we get**

$$\begin{aligned} u^2\cos^2\theta &= \frac{2}{3}\left(u^2 - \frac{u^2\sin^2\theta}{2}\right) \\ \Rightarrow 3\cos^2\theta &= 2 - \sin^2\theta \\ \Rightarrow 3(1-\sin^2\theta) &= 2 - \sin^2\theta \\ \Rightarrow \sin^2\theta &= \frac{1}{2} \\ \Rightarrow \sin\theta &= \frac{1}{\sqrt{2}} \end{aligned}$$

which gives $\theta = 45^\circ$**. So the correct choice is (b).**

Q.20 Answer:

$\frac{u}{2}\left(1+3\cos^2\theta\right)^{1/2}$

Solution:

Refer to Fig. is the highest point on the trajectory.

$$\begin{aligned} \text{Average velocity} &= \frac{\text{displacement } OA}{\text{time taken}} \\ &= \frac{\sqrt{h_{\max}^2 + \frac{R^2}{4}}}{\frac{1}{2}t_f} \\ \text{where} \quad h_{\max} &= \frac{u^2\sin^2\theta}{2g} \\ R &= \frac{u^2\sin^2 2\theta}{g}, \end{aligned}$$

and $t_f = \frac{2u\sin\theta}{g}$

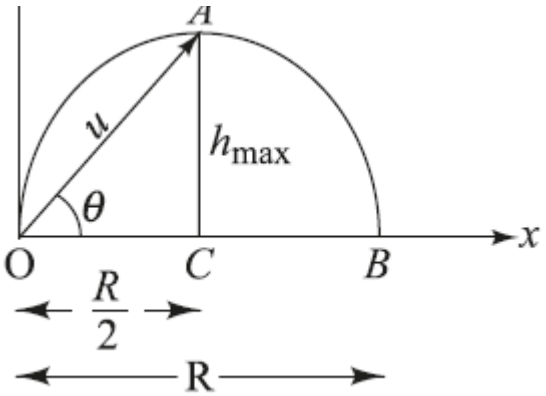
Using these in Eq. (1) and simplifying, we get

Average velocity $= \frac{u}{2}\left(1+3\cos^2\theta\right)^{1/2}$

So the correct choice is (d).



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